



JC3008 Knowledge Representation

Day 1: First Order Logic

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Day 1

First Order Logic

1. Introduction
2. Syntax
3. Semantics
4. Knowledge Base

Motivation

Why do we need to use a logic to represent knowledge? Can't we just use English (or any other natural language?)

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Why do we need to use a logic to represent knowledge? Can't we just use English (or any other natural language?)

- Creating an inference engine for a natural language is hard
- Natural languages are full of ambiguity
- Formal logics on the other hand can be as expressive as natural languages and at the same time are equipped with the tools to reason about sentences
- Logical languages study the use of entailment relations and rules of inference to arrive at truth conditions

Applications

Are these logics used anywhere in practice?

- Formal verification
- Legal reasoning
- Mathematical proofs
- Knowledge bases

KR&R and Logic

- Knowledge Representation: Symbolic encoding of propositions believed (by some agent)
- Logic: A tool for “symbolic language” to represent knowledge
- Start with First-order predicate calculus (FOL)
 - Invented by philosopher Frege for the formalisation of mathematics but will consider subsets / supersets and very different looking representation languages.
- Why FOL and not propositional logic?

KR&R and Logic

- Knowledge Representation: Symbolic encoding of propositions believed (by some agent)
- Logic: A tool for “symbolic language” to represent knowledge
- Start with First-order predicate calculus (FOL)
 - Invented by philosopher Frege for the formalisation of mathematics but will consider subsets / supersets and very different looking representation languages.
- Why FOL and not propositional logic?
 - Simple, serves as a good introduction to KRR
 - Expressive

Propositional Logic

- Concerned with propositions and their interrelationships
- Roughly speaking, a *proposition* is a possible condition of the world about which we want to say something. In other words, something that can be evaluated to true or false.
 - Proposition A: “Earth is round” (evaluated to *True*)
 - Proposition B: “Water contains carbon atoms” (evaluated to *False*)
- p, q are propositions so are $\neg p, p \wedge q, p \vee q, p \Rightarrow q (\neg p \vee q)$ and $p \Leftrightarrow q$
- p, q are propositional variables, but $\neg p, p \wedge q, p \vee q, p \Rightarrow q (\neg p \vee q)$ and $p \Leftrightarrow q$ are compound variables.

Propositional Logic

- Logical reasoning methods are designed to work no matter what meanings or values are assigned to the logical "variables" used in sentences. (ie. P by itself could be true or false).
- Formally, an *interpretation* for propositional logic is a mapping assigning a truth value to each *propositional variable* of the language.
- Example: for propositions p, q, r we define interpretation i such that $p^i = \text{true}$, $q^i = \text{false}$, $r^i = \text{true}$, then $(p \Rightarrow q)$ is evaluated to false.
- Satisfiability: We say that an interpretation i satisfies a sentence (e.g. propositional or compound variables) if and only if it is true under that interpretation.
- Validity: A sentence is *valid* if and only if it is satisfied by every interpretation. (ex. $p \vee \neg p$)

First Order Logic

- Propositional Logic gives us the machinery to derive logical conclusions based on the relationship between propositions but this is not adequate.
- Relational Logic (or FOL), provides us with a way of talking about individual objects and their interrelationships. It allows us to assert the existence of objects with a given relationship and allows us to talk about all objects having a given relationship.
- Also known as Predicate Logic or Predicate Calculus (application of predicate symbols to terms which evaluate to true or false)
- Very expressive
 - “Socrates is a man”
 - “There exists x such that x is Socrates and x is a man”
- Two main elements: Syntax (how to write) vs Semantics (how to interpret)

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Alphabet

Logical	Symbols
punctuation	$() [] . ,$
connectives ¹	$\wedge$ (conjunction) $\vee$ (disjunction) $\rightarrow$ (implication) $\Leftrightarrow$ (biconditional) $\neg$ (negation) $=$ (equality)
quantifiers variables	$\forall$ (universal quantification) $\exists$ (existential quantification) $x, x_1, x_2, \dots, y, z$
Non-logical	Symbols
predicates functions	$P(x)$ e.g., $\text{Lecturer}(F)$ (uppercase) $f(x)$ e.g., $\text{teaching}(\text{ferdi}, \text{jc4002})$ (lowercase)

- Non-logical symbols have arity (number of arguments)
 - 0-arity predicates: propositional variables (e.g., F in $\text{Lecturer}(F)$)
 - 0-arity functions: constants (e.g., ferdi , jc4002 in $\text{teaching}(\text{ferdi}, \text{jc4002})$)

¹You need to know truth tables, for example see [here](#).

Grammar

Terms

1. Every variable is a *term*.
2. If $t_1, t_2, \dots, t_n$ are *terms* and f is a function of arity n , then $f(t_1, t_2, \dots, t_n)$ is a *term*

Atomic wffs (well-formed formula)

- 1. If $t_1, t_2, \dots, t_n$ are *terms* and P is a predicate of arity n , then $P(t_1, t_2, \dots, t_n)$ is an *atomic wff*.
- 2. If t_1 and t_2 are terms, then $(t_1 = t_2)$ is an *atomic wff*.

Wffs

- 1. Every *atomic wff* is a *wff*.
- 2. If α and β are *wffs*, and v is a *variable*, then $\neg\alpha$, $(\alpha \wedge \beta)$, $(\alpha \vee \beta)$, $(\alpha \rightarrow \beta)$, $\forall v \alpha$, $\exists v \alpha$ are *wffs*.

A sentence: wff with no free variables (closed)

Activity 1

In pairs, discuss the following question (5 minutes) and then share your answers with the rest of your colleagues (5 minutes):

“In your opinion, what are the differences between PL and FOL?”

Variable Scope

Example FOL formula:

$$\frac{\frac{}{\forall x P(f(x), x) \rightarrow \exists y (P(f(g(x, y)), g(x, y)) \wedge Q(x, f(x)))}}{1}}{2}$$

- The scope of $\forall x$ is 1
- The scope of $\exists y$ is 2
- Are all variables **bound** to a quantifier?

Variable Scope

Example FOL formula:

$$\overline{\overline{\forall x P(f(x), x) \rightarrow \exists y (P(f(g(x, y)), g(x, y)) \wedge Q(x, f(x)))}}_2}_1$$

- The scope of $\forall x$ is 1
- The scope of $\exists y$ is 2
- Are all variables **bound** to a quantifier? (yes)
- If at the end of the formula we had $\wedge r(y)$, then this variable would be free
- *FOL sentence*: well-formed formula with no free variables

Well-Formed Formula

Example FOL formula:

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Is this a well-formed formula?

Well-Formed Formula

Example FOL formula:

$$\frac{\frac{}{1}}{\frac{}{2}} \quad \forall x P(f(x), x) \rightarrow \exists y (P(f(g(x, y)), g(x, y)) \wedge Q(x, f(x)))$$

Is this a well-formed formula?

- First we apply quantifier rules from wff definition to the quantifiers we have in the formula $\forall x\phi$ and $\exists y\psi$
- Then we apply the implication rule from wff definition to $\phi \rightarrow \psi$
- We apply conjunction rule in ψ , which becomes $\alpha \wedge \beta$
- Finally, we transform ϕ , α , β to predicate symbols.

From English to FOL

- Alice is a student
- There is a student

From English to FOL

- Alice is a student
 $Student(Alice)$
- There is a student
 $\exists x \text{ } Student(x)$
- There is a student that has not colluded with another student

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 $\exists x (Student(x) \wedge \forall y (Student(y) \wedge \neg(x = y) \rightarrow \neg Colluded(x, y)))$
- Every student takes at least one module

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$\forall x (Student(x) \rightarrow \exists y (Module(y) \wedge Takes(x, y)))$

- Alice takes either KRR or ML (but not both)

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- Alice takes either KRR or ML (but not both)

$Takes(Alice, KRR) \Leftrightarrow \neg Takes(Alice, ML)$

Precedence of Logical Operators

1. Negation $\neg$
 2. Conjunction $\wedge$
 3. Disjunction $\vee$
 4. Quantifiers $\forall$ and $\exists$
 5. Implication $\rightarrow$
 6. Biconditional $\Leftrightarrow$
- Operators with the same precedence are evaluated left to right
 - Extra punctuation (such as parenthesis) that are not required can be added to improve readability of a formula

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Semantics

- How do we interpret FOL sentences?
 - Logical symbols have fixed meaning, however this is not the case for non-logical symbols
 - These non-logical symbols may reach outside the specified language
 - We need a way to interpret these symbols
 - A specification of how to understand predicate and function symbols
- Enter logical interpretations

Interpretations

- Interpretation $\zeta = (D, I)$, where D is the domain of discourse (non-empty set, contains entities for which certain variables may range) and I is the interpretation function
- An interpretation function is assigned to each non-logical symbol
 - Each function symbol f of arity n gets a function $I(f)$ from D^n to D
 - Each predicate symbol P of arity n gets a function $I(P)$ from D^n to $\{\text{true}, \text{false}\}$
- Thus, an interpretation assigns a denotation (from terms to elements of the domain) to each non-logical symbol
- Useful to clear up any ambiguities from non-logical symbols

Satisfaction

- How can we use this to evaluate the truth values of a formula?
- Given an interpretation and a variable assignment (μ), a formula evaluates to either true or false
 - μ associates an element of the domain with each (free) variable
- By inductive definition we can use the truth schema:
 - Atomic formulae: depends on the evaluation of all the individual terms
 - Logical connectives: follow the truth table for the respective connective
 - Existential quantifier: $\exists x \phi(x)$, only true if there is a way to select a value for x such that $\phi(x)$ is satisfied
 - Universal quantifier: $\forall x \phi(x)$, only true if every possible choice of a value for x causes $\phi(x)$ to be satisfied

Interpretation Example

Given the domain of discourse $D = \{0, 1, 2\}$ and the interpretation mappings:

- $I(P) = \{0, 1, 2\}$
- $I(Q) = \{(0, 0), (1, 1), (2, 2)\}$
- $I(R) = \{(0, 1), (2, 0), (1, 2)\}$

Indicate whether the following formulae are true or false and provide a justification.

- $P(0, 1, 2)$

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- $P(0, 1, 2)$ false, P can only have one element
- $P(3) \rightarrow P(0)$

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Indicate whether the following formulae are true or false and provide a justification.

- $P(0, 1, 2)$ false, P can only have one element
- $P(3) \rightarrow P(0)$ true, implication truth table, left side is false and right side is true, which resolves to true
- $P(1) \wedge Q(1, 1) \wedge R(0, 1)$

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- $P(1) \wedge Q(1, 1) \wedge R(0, 1)$ true, each of the 3 parts are true, therefore the conjunction is true
- $\forall x \exists y P(x) \wedge Q(x, x) \wedge R(y, x)$

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- $P(1) \wedge Q(1, 1) \wedge R(0, 1)$ true, each of the 3 parts are true, therefore the conjunction is true
- $\forall x \exists y P(x) \wedge Q(x, x) \wedge R(y, x)$ true, x can be 0, 1, or 2, and for every x there is a y (either 0, 1, or 2) that makes the formula true

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Steps for (FOL) Knowledge Engineering

1. Identify the task
2. Assemble the relevant knowledge
3. Decide on a vocabulary of predicates, functions, and constants
4. Encode general knowledge about the domain (rules)
5. Encode a description of the specific problem instance
6. Pose queries to the inference procedure and get answers
7. Debug the knowledge base

Example in Natural Language

1. Alice is a professor
2. All professors are people
3. John is the dean
4. Deans are professors
5. All professors consider the dean a friend or don't know him
6. Everyone is a friend of someone
7. People only criticise people that are not their friends
8. Alice criticised John
9. Is John a friend of Alice?

Translating to FOL (Human Readable)

1. Prof (Alice)
2. $\forall x \text{ (Prof (x) } \rightarrow \text{ Person(x))}$
3. Dean(John)
4. $\forall x \text{ (Dean(x) } \rightarrow \text{ Prof (x))}$
5. $\forall x \exists y \text{ (Prof (x) } \wedge \text{ Dean(y) } \rightarrow \text{ Friend (x, y) } \vee \neg \text{Knows(x, y))}$
6. $\forall x \exists y \text{ Friend (x, y) }$
7. $\forall x \forall y \text{ (Person(x) } \wedge \text{ Person(y) } \wedge \text{ Criticise(x, y) } \rightarrow \neg \text{Friend (x, y))}$
8. Criticise(Alice, John)
9. Is this query true or false? $\neg \text{Friend (John, Alice) }$

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5. $\forall x \exists y \text{ (Prof (x) } \wedge \text{ Dean(y) } \rightarrow \text{ Friend (x, y) } \vee \neg \text{Knows(x, y))}$
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 - John is a person; sentences 3, 4, and 2

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 - John is a person; sentences 3, 4, and 2
 - Alice is a person; sentences 1 and 2

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8. Criticise(Alice, John)
9. Is this query true or false? $\neg \text{Friend (John, Alice)}$
 - John is a person; sentences 3, 4, and 2
 - Alice is a person; sentences 1 and 2
 - Sentence 7 fails because John has not criticised Alice
 - Query result is false

Activity 2

In pairs, discuss the following question (5 minutes) and then share your answers with the rest of your colleagues (5 minutes):

“In your opinion, why does the previous problem fail? How can we make it work?”

Translating to FOL (Machine Readable)

1. $P1(A)$
2. $\forall x (P1(x) \rightarrow P2(x))$
3. $P3(B)$
4. $\forall x (P3(x) \rightarrow P1(x))$
5. $\forall x \exists y (P1(x) \wedge P3(y) \rightarrow P4(x, y) \vee \neg P5(x, y))$
6. $\forall x \exists y P4(x, y)$
7. $\forall x \forall y (P2(x) \wedge P2(y) \wedge P6(x, y) \rightarrow \neg P4(x, y))$
8. $P6(A, B)$
9. Query: $\neg P4(B, A)$

Summary

- Discussed the benefits of logic languages, specifically FOL
- Learned the syntax and semantics of FOL
- How to use FOL to make a knowledge base

Until the next session:

- Install SWI-Prolog to your computer.